2(a) $p(x|\mu) = \frac{\mu^x e^{-\mu}}{x!}$, $x \in \mathbb{Z}$

$\mu = \lambda t$

$\Rightarrow p(k|\lambda) = \frac{(\lambda t)^k e^{-(\lambda t)}}{k!}$

(b) $\langle k \rangle = \sum_{k=1}^{\infty} k\, p(k|\lambda)$

$= \sum_{k=1}^{\infty} \frac{(\lambda t)^k k}{k!} e^{-(\lambda t)}$

$= (\lambda t) e^{-\lambda t} \sum_{k=1}^{\infty} \frac{(\lambda t)^{k-1}}{(k-1)!}$

$= (\lambda t) e^{-\lambda t} \sum_{n=0}^{\infty} \frac{(\lambda t)^n}{n!}$

$= \lambda t$

(c) $\langle k^2 \rangle = \sum_{k=1}^{\infty} k^2 p(k|\lambda)$

$= \sum_{k=1}^{\infty} \frac{(\lambda t)^k k^2}{k!} e^{-(\lambda t)}$

$= (\lambda t) e^{-\lambda t} \sum_{k=1}^{\infty} \frac{(\lambda t)^{k-1} k}{(k-1)!}$

$= (\lambda t) e^{-\lambda t} \left[\sum_{k=2}^{\infty} \frac{(\lambda t)^{k-1}}{(k-1)!}(k-1) + \sum_{k=1}^{\infty} \frac{(\lambda t)^{k-1}}{(k-1)!} \right]$

$= (\lambda t)^2 e^{-\lambda t} \sum_{n=0}^{\infty} \frac{(\lambda t)^n}{n!} + \lambda t$

$= (\lambda t)^2 + \lambda t$

(d) $\mathrm{Var}(k) = \langle k^2 \rangle - \langle k \rangle^2 = \lambda t = \langle k \rangle$

4(a) $p(\{x_n\} | \mu, \sigma^2)$

$$= \prod_{n=1}^{N} \mathcal{N}(x_n | \mu, \sigma^2)$$

(b) $\ln p(\{x_n\} | \mu, \sigma^2)$

$$= \frac{-1}{2\sigma^2} \sum_{n=1}^{N} (x_n - \mu)^2 - \frac{N}{2} \ln \sigma^2 - \frac{N}{2} \ln(2\pi)$$

(c) $$\frac{\partial}{\partial \mu} \ln p \Big|_{\mu = \mu_{ML}} = 0$$

$$\Rightarrow \mu_{ML} = \frac{1}{N} \sum_{n=1}^{N} x_n$$

(d) $$\frac{\partial}{\partial \sigma} \ln p \Big|_{\sigma = \sigma_{ML}} = 0$$

$$\Rightarrow \sigma_{ML}^2 = \frac{1}{N} \sum_{n=1}^{N} (x_n - \mu_{ML})^2$$

(e) $$\sigma^2_{unbiased} = \frac{N}{N-1} \sigma^2_{ML}$$

$$= \frac{1}{N-1} \sum_{n=1}^{N} (x_n - \mu_{ML})^2$$

Bias comes from estimating variance from the sampled mean μ_{ML} instead of true mean

Relevant when $N \to \infty$.

5 $\quad \mu_{N+1} - \mu_N = \delta = \dfrac{x_{N+1} - \mu_N}{N+1}$

$$\Sigma^{N+1} = \frac{1}{N+1} \sum^{N+1} (x_n - \mu^{N+1})(x_n - \mu^{N+1})^T$$

$$= \frac{1}{N+1} \sum^{N} (x_n - \mu^N - \delta)(x_n - \mu^N - \delta)^T$$

$$+ \frac{1}{N+1} (x_{N+1} - \mu^N - \delta)(x_{N+1} - \mu^N - \delta)^T$$

$$= \frac{1}{N+1} \sum_{n=1}^{N} [(x_n - \mu^N)(x_n - \mu^N)$$

$$- \delta \underbrace{(x_n - \mu^N)^T}_{=0} - \underbrace{(x_n - \mu^N)}_{=0} \delta^T + \delta\delta^T)$$

$$+ \frac{1}{N+1} [(N+1)\delta - \delta][(N+1)\delta - \delta]^T$$

$$= \frac{N}{N+1} (\Sigma^N + \delta\delta^T) + \frac{N^2}{N+1} \delta\delta^T$$

$$= \frac{N}{N+1} \Sigma^N + N\delta\delta^T$$